Questions

What are the political possibilities (and limits) of ‘artistic critique’? How should we understand it in relation to the broader tradition of AI critique in the humanities and social sciences?

Where are the openings in today’s ‘inhospitable institutional and economic conditions’, inside which artists might approach contemporary AI with a ‘sense of possibility’?

What factors might make the third AI winter different?

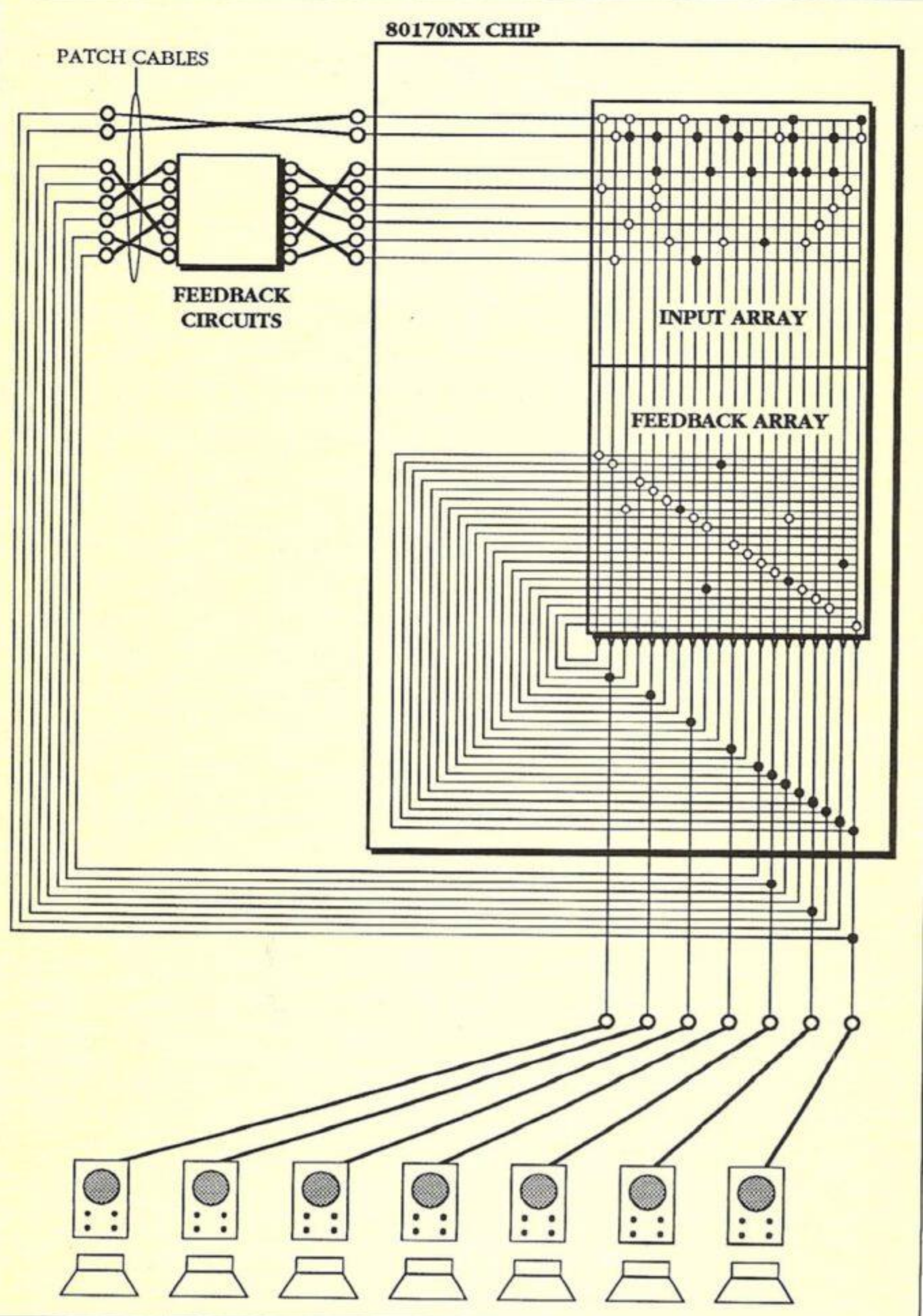
AI Winters, Experimental Music Springs

AI Critique in the Music of Maryanne Amacher and David Tudor

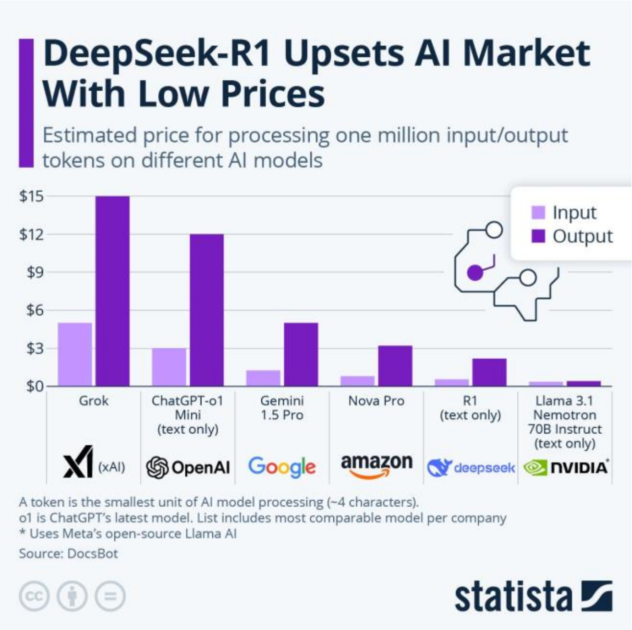
MusAI pre-NIME workshop, 20 June 2026

Christopher Haworth (University of Birmingham)

Music and Artificial Intelligence: Building Critical Interdisciplinary Studies
Grant agreement ID: 101019164



What is an AI winter?

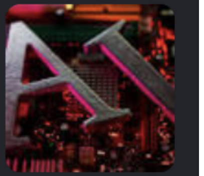


The Telegraph

The next 'AI winter' is coming

Artificial Intelligence turns 70 years old next year, but the birthday celebrations may be muted. The stock market is braced for a crash: a...

24 Aug 2025



Fortune

Is another 'AI winter' coming? Here's what past AI slumps can tell us about what the future might hold

As summer fades into fall, many in the tech world are worried about winter. Late last month, a Bloomberg columnist asked "is the AI winter...

3 Sept 2025

The Guardian

OpenAI shutters AI video generator Sora in abrupt announcement

Tech firm 'says goodbye' to Sora, made publicly available in 2024, just six months after its launch of a stand-alone app.

25 Mar 2026

What is an AI winter?

Drew McDermott:
In spite of all the commercial hustle and bustle around AI these days, there's a mood that I'm sure many of you are familiar with of deep unease among AI researchers who have been around more than the last four years or so. This unease is due to the worry that perhaps expectations about AI are too high, and that this will eventually result in disaster.

To sketch a worst case scenario, suppose that five years from now the strategic computing initiative collapses miserably as autonomous vehicles fail to roll. The fifth gen-

what extent is it due to näiveté on the part of the public? What is the role of the press in this mismatch, and how can we help to make the press a better channel of communication with the public? What is the role of funding agencies in the future going to be as far as keeping a realistic attitude toward AI? Can we expect DARPA and ICOT to be stabilizing forces, or is there a danger that they may cause people in government and business to get a little bit too excited? Are funding agencies going to continue to fund pure research, even if AI becomes a commercial success? Will the perception remain that we need to do

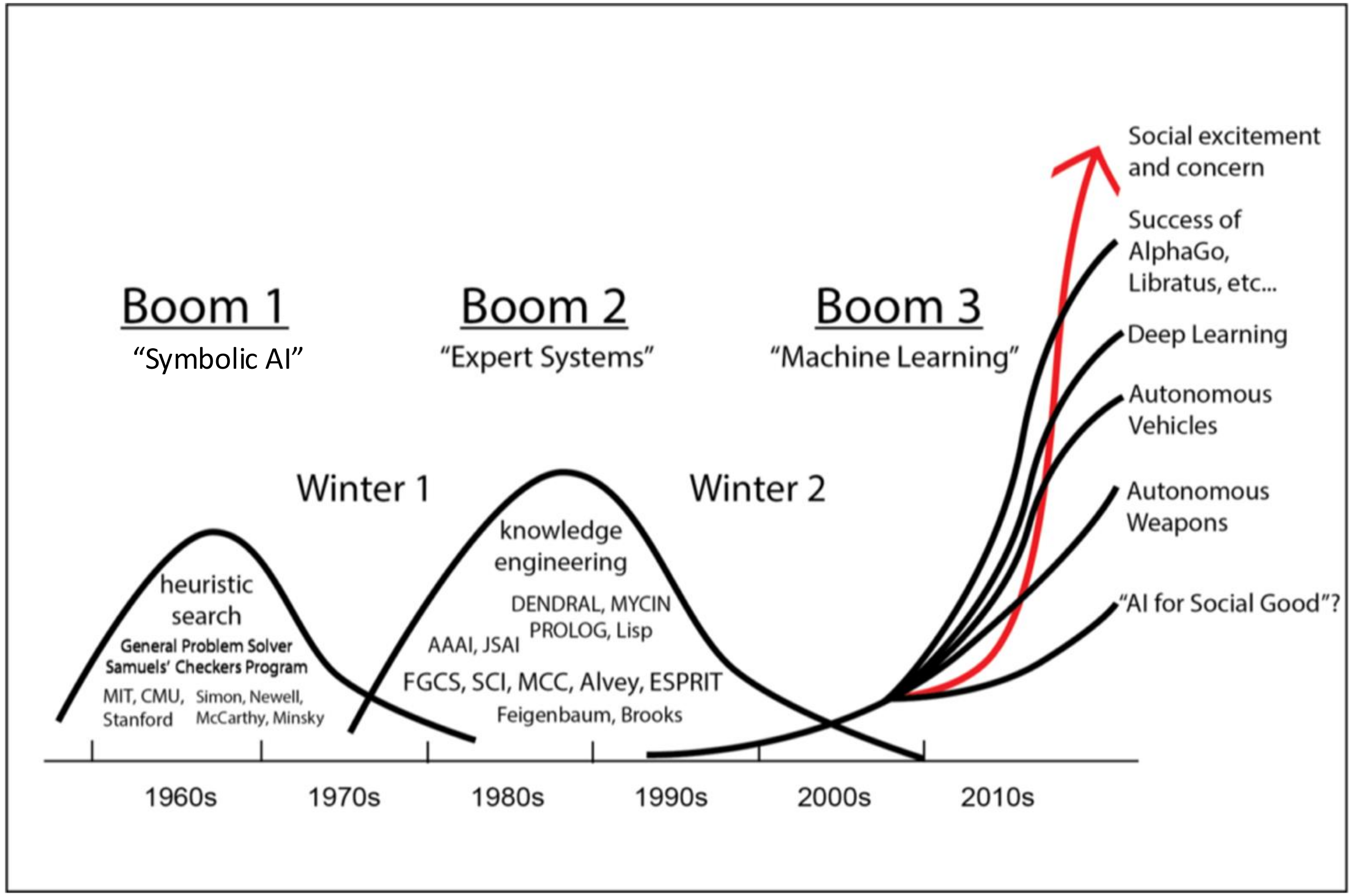
To sketch a worst case scenario, suppose that five years from now the strategic computing initiative collapses miserably as autonomous vehicles fail to roll. (…) Every startup company fails. (…) And there's a big backlash so that you can't get money for anything connected with AI. (…) This condition, called the 'AI winter' by some, prompted someone to ask me if the 'nuclear winter' were the situation where funding is cut off for nuclear weapons.

This panel has been assembled to discuss these issues. I've asked the panelists to discuss the following questions in particular: Are expectations too high among consumers of AI, such as business and military? If they are too high, then why? Is there something we can do to change this mismatch between expectation and reality? To what extent is this mismatch our fault? There's a charge often leveled against AI people that they claim too much. To

of several days, work up a human impact statement for artificial intelligence. Marvin Minsky, as well as myself and others, were on the resource panel. The result wa
what you might expect: A combination of silliness a
seriousness, with not a great deal of informed insight ir
AI. But there was a very good cross-section of the gene
public, and I gained some very interesting insights wh
trying to answer their questions.

McDermott, D., Waldrop, M.M., Chandrasekaran, B., McDermott, J. and Schank, R., 1985. The dark ages of AI: a panel discussion at AAAI-84. *AI Magazine*, *6*(3), pp.122-134.

What is an AI winter?



Two Provocations

Provocation 1: Notion of the AI winter can be used rethink certain topics in the political-economic and political-aesthetic history of experimental music

Two Provocations

Provocation 1: Notion of the AI winter can be used rethink certain topics in the political-economic and political-aesthetic history of experimental music

Provocation 2: Experimental music has also been a key resource for *AI critique* over its several waves of enthusiasm and investment

Part one: Maryanne Amacher and Marvin Minsky at MIT

The Artificial Intelligence Group at MIT, 1963-1970

-Hearing Scientist and MIT lecturer J.C.R. Licklider heads Information Processing Techniques Office of Advanced Research Projects Agency (ARPA), 1962-64.

-DARPA funds AI research at Stanford, Carnegie Mellon, Illinois Urbana Champain, MIT, Harvard, also commercial R&D at IBM. Computing research at MIT receives $3-4 mil annually. 'I don't recall ever writing a proposal for that' –M. Minsky

-MIT's Project MAC (dir. Licklider) / Artificial intelligence Group (dir. Minsky)

-Project MAC developed computer systems, computer languages, graphical user interfaces, and other utilities like compilers and debuggers

-AI group aimed to understand and reproduce human cognition through computer software emulations

Allen Newell and Herbert A. Simon AI prediction (1957)

“Within ten years:

-a computer will be world chess champion

-a computer will discover and prove an important unknown mathematical theorem

-most psychological theories will take the form of computer programs

-a computer will compose aesthetically valuable music”

Marvin Minsky

Minsky a founder of AI as interdisciplinary field marrying engineering, cognitive science, and rationalist philosophy

Also a gifted classical pianist, he saw music as an exemplary field for exploration of artificial intelligence

Used music to exemplify problems of cognition, like learning, entrainment, and goal orientation

‘The modern tendency, to be tolerant and say “anything is music,” is bad for the mind, both of the listener and of the critic also’ (Laske 1992, 34).

‘Music’ for Minsky synonymous with the ‘problem space’ of the equal tempered keyboard, and with the humanly-limited, two-handed practice of polyphonic counterpoint.

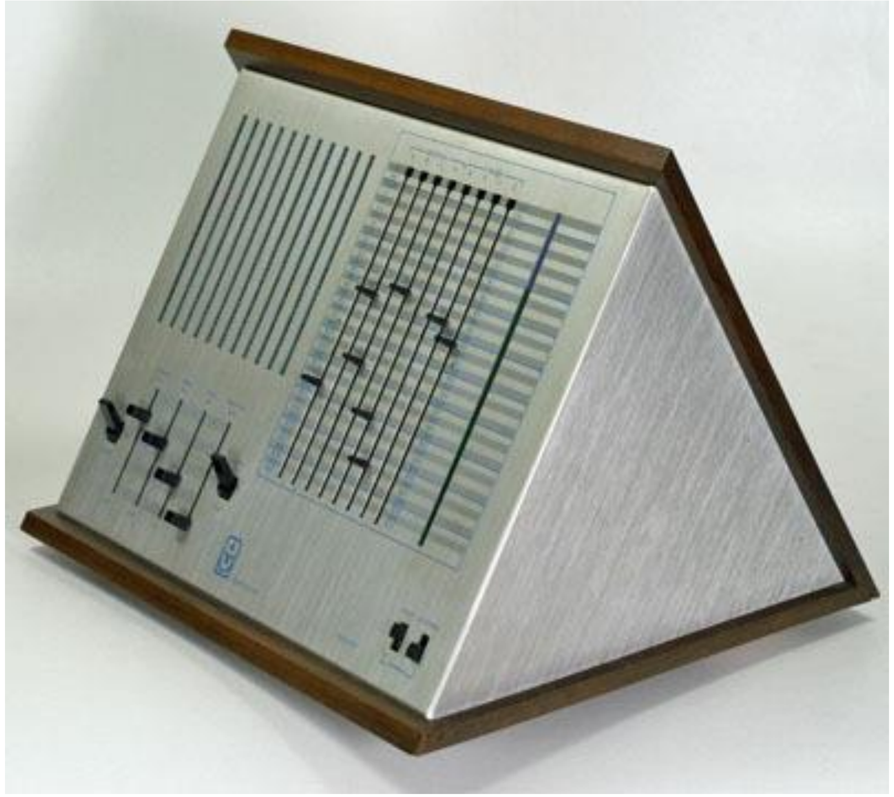
Triadex Muse (Marvin Minsky & Edward Fredkin)

Minsky develops the Triadex Muse system in collaboration with Edward Fredkin in 1969, commercialised in 1970

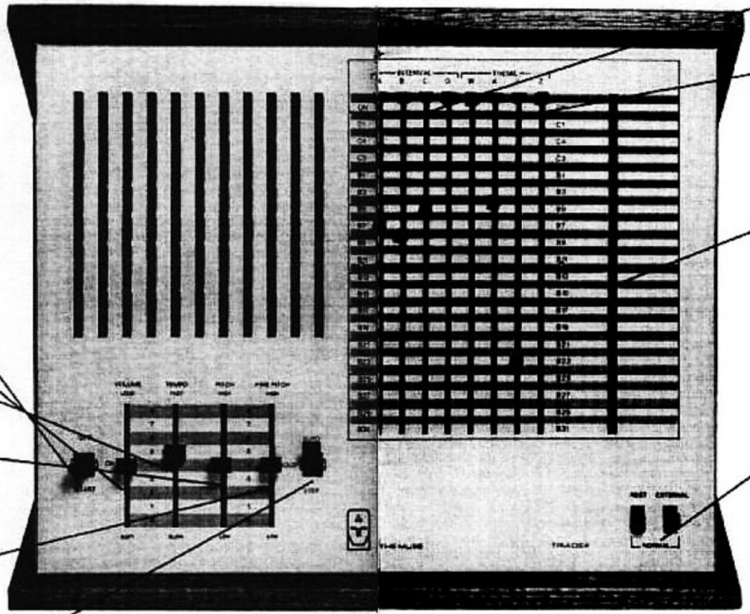
‘Intelligent’ instrument that uses bit shifting matrix to generate non-repeating note patterns

Made use of the new digital integrated circuits that came on the market in the late 60s, and which allowed for basic logic programming, counting, and bit-shifting operations

Device stems from the ‘appreciation that a very high degree of randomness is not necessary (…) facilitates the construction of relatively simple apparatus which nonetheless automatically generates relatively pleasing music’ (1970, 1).



Triadex Muse (Marvin Minsky & Edward Fredkin)



Conductor Section

Off-On-Start. When you press down to Start position, the composition goes back to the very beginning.

Volume. Upward for loud, down for soft.

Tempo. Upward for very fast, down for slow.

Pitch. Upward for high soprano, down for bass. If you move the Pitch switch rapidly while the Muse is playing, you can get many unusual siren effects. Moving it slowly while the Muse is playing is a good way to get most people to leave the room.

Fine Pitch. This is for making delicate adjustments of pitch. Mainly for use when playing the Muse with other instruments.

Auto-Hold-Step. Should normally be up in the Auto position for general use. Placed in the center or Hold position, the Muse will instantly stop wherever it is in the music. By pressing it down to the Step position, the Muse will play one note at a time. Note: try pressing down to Step many times. You are now controlling the Muse's playing by yourself, which means you can play any rhythm you can "tap" on the switch.

Composer Section

Interval. Switches A, B, C, and D. These control the notes to be used in a particular composition.

Theme. Switches W, X, Y, and Z. These switches, particularly when placed in the B-regions (corresponding to the green lights) create complex and intricate variations on a basic melody.

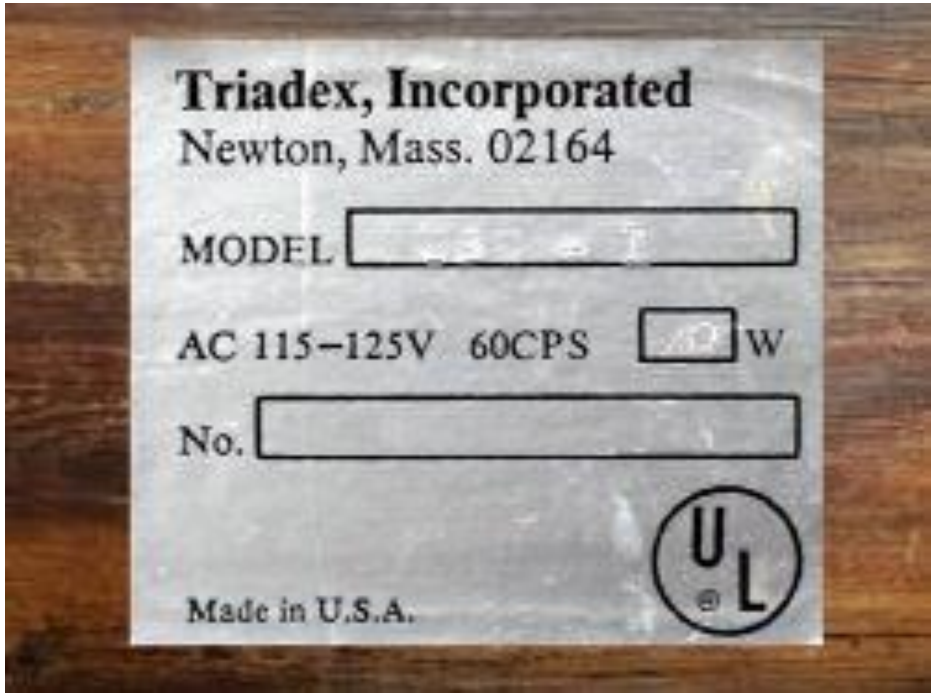
Indicator lights, blue and green. The lights represent the three regions: Off/On, the C region (blue), and the B region. (green) Each row, and its light, represents one position for the Composer switches. The various positions are *off* or *on* as indicated by the lights being *off* or *on*. Any Composer switch will be *on* at any given moment when its corresponding light is on.

Rest and External Switches. The Rest switch, when in the up or Rest position, introduces silent pauses in the development of your composition, in place of the lowest note. Many, but not all, tunes are improved by this.
The External switch should be kept in the Normal position.

Triadex Muse (Marvin Minsky & Edward Fredkin)

1) Muser’s Waltz | Themes and Variations (1970) by Marvin Minsky

Triadex Muse (Marvin Minsky & Edward Fredkin)



Maryanne Amacher at MIT 1972-76

Amacher a research fellow at MIT's Center for Advanced Visual Studies ("CAVS") between 1972 and 1976

Produces 'City Links' pieces in accord with CAVS's civic mission of raising awareness about environmental remediation at the Charles river in Boston

Generation of composers who reacted against fixity and finitude of recording media. Wanted media to 'match the sensitive range of our responsive energies'.

Collaborates with Minsky on a series of Muse+Computer experiments

Produces 'head tone' music using the Muse

Maryanne Amacher at MIT 1972-76

Head Rhythm 1 / Plaything 2 (1999) by Maryanne Amacher

“Head Rhythm 1" offers starry tone bursts layered in charmingly straightforward patterns. This character's overall feel is quartal. From both the left and right channels, dyads alternate in short, five- pulse patterns punctuated by an anacrustic pause. But ‘head rhythm’ also plies the stereo spread’s outer edges. In the right channel, the layers reach four octaves above middle c, with eight and sixteenth note patterns blazing by at about 140 to the quarter note. The slower left channel shares its lowest stratum with the right’s, and it adds a stepwise line that flits between its higher layers at 105. “Head Rhythm 1” spins through the room in a three-against-four pattern, crating a texture pocked with negative space. As clicks and pops proliferate, something slightly disconcerting might begin to fill the space between the tone bursts. One’s inner ear perhaps starts to feel warm; a little buzz in the ear catches one’s attention. Individual frequencies come into focus, short gestures and melodies perhaps coalesce. These are what Amacher called ‘ear tones’, and on this record, perhaps a ‘third ear’

Amy Cimini, 2022. Wild Sound: Maryanne Amacher and the Tenses of Audible Life, p.170.

Species of critique?

-creative ‘misuse’ in use of Triadex Muse

-pro-cybernetic listening in emphasis on situational effects

-critique of AI’s human supremacism (‘intelligence’) via inter-mammalian otoacoustic emissions (‘life’)

Part two: David Tudor and Intel Computers, 1989-1993

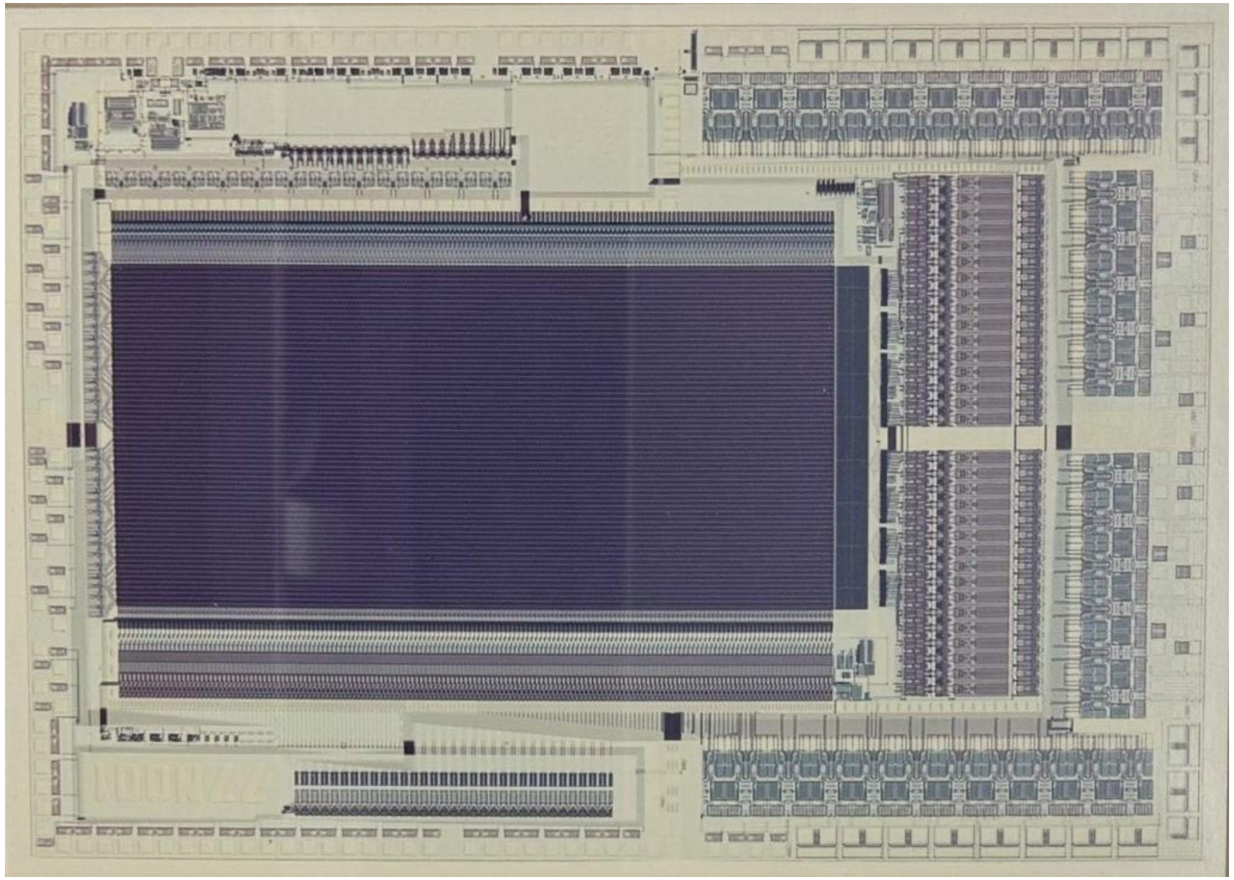
Tudor a key experimental musician known as pianist/interpreter of Euro-American avant-garde and pioneer of DIY electronics

At end of his career, engineer Forest Warthman enrols Mark Holler of Intel propose to develop a ‘universal’ system for Tudor, to rationalise his boxes and wires

‘”*I was struck by the fact there were so many different kinds of devices going. At an intermission, I asked David if he'd be interested in some electronic device that combined all of those sources that he could control, one device that had all of the sources. And he said, yes, he would be quite interested*” (interview with author, 2023)

List of artist consultants provided by Tudor to help design system: Billy Kluver, Nick Collins, Serge Tcherepnin, La Monte Young, Max Neuhaus, David Ziccarelli…

David Tudor and Intel, 1989-1993



80170NX Electrically Trainable Neural Network chip (ETANN)

David Tudor and Intel, 1989-1993

CHM Computer History Museum

Hours & Admission | Subscribe | Account

EXPLORE | VISIT | ABOUT | JOIN & GIVE | DONATE | TICKETS

HOME › CHM BLOG › FROM THE COLLECTION

NEURAL NETWORK CHIP JOINS THE COLLECTION

By Dag Spicer | August 09, 2024

David Tudor and Intel, 1989-1993

Real-Time Pattern Recognition for Embedded Neural Networks

Advanced IC Technology for Pattern Recognition Applications – Today

Intel's 80170NX Electrically Trainable Analog Neural Network (ETANN) chip offers an entirely new way to solve a variety of pattern recognition problems in real time. The 80170NX ETANN combines advanced IC technology with a high-density parallel architecture to deliver greater throughput per dollar than alternative technologies.

A complete set of easy-to-use hardware and software tools for simulating, training, and prototyping, plus expert support from Intel's engineering staff, comprise Intel's total solution for neural network applications. And it's available today, at a remarkably low cost.

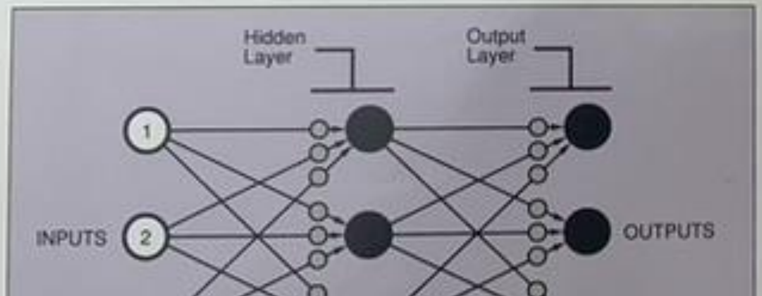
The 80170NX ETANN performs complex neural network functions at extremely high speed, mapping input data into decision-oriented outputs. Its integration and low-power requirements make it a high-performance embedded solution for compact, low-power systems that implement small- to medium-sized, multi-layer neural networks.

Pattern recognition and signal processing applications are prime candidates for 80170NX-based systems. Continuous processes can be modeled, so as to forecast

Here are some of the specific application areas to which the 80170NX is well suited:

- *Signal Processing* – Image processing, machine vision, machine-failure diagnostics, medical diagnostics, sonar and ultrasound recognition, target detection, adaptive filters, signal compression, vector quantization.
- *Process Optimization* – Vehicle control, quality control, modeling of continuous processes, dynamic process planning and scheduling.
- *Robotic Motion* – Inverse kinematic transformations, coordinate transformations, combined sensor-motor control, sensor fusion, balancing.
- *Associative Memory* – Content-addressable memory, auto correlation, error correction.

The 80170NX is best suited to applications where pattern recognition tasks are stable, since the chip is trained on a separate training system. This makes it possible to use a wide variety of software-based training algorithms, several of which are currently offered and supported by Intel.



The Intel Edge

Real-Time Speed

A single 80170NX chip can perform more than 2 billion multiply-accumulate operations (connections) per second. By interconnecting eight chips, systems can achieve more than 16 billion connections per second, a performance level that exceeds most supercomputers.

Multiple Learning Algorithms

You can train the 80170NX using a wide variety of methods, including back-propagation, recurrent back-propagation, Madaline Rule III, and your own custom learning algorithms. Intel's Neural Network Training System and third-party simulation software support these methods.

Nonvolatile Storage

The 80170NX chip is fabricated with Intel's CHMOS III EEPROM technology,

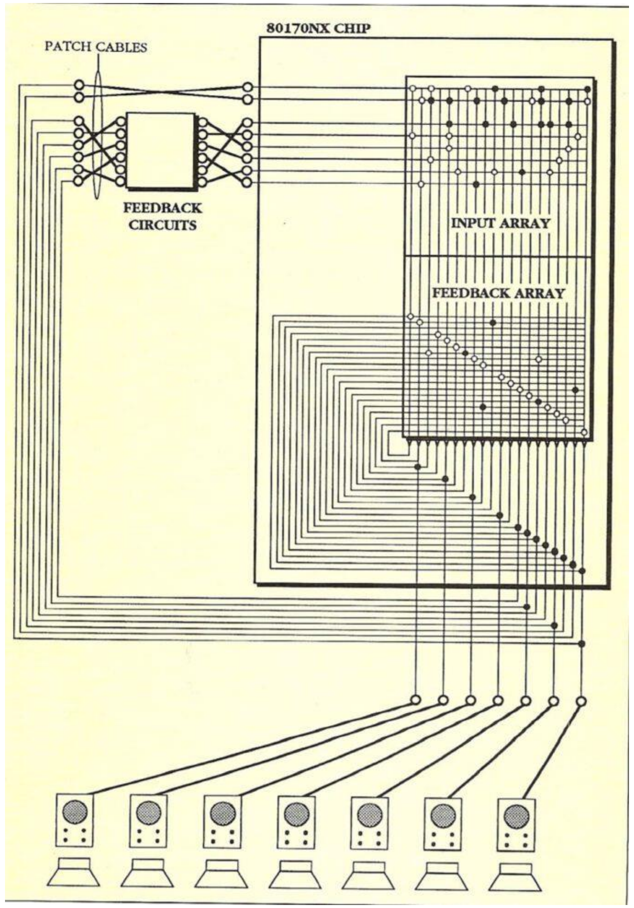
"…it was a technology looking for an application"

-Mark Holler (2023, Interview with author)

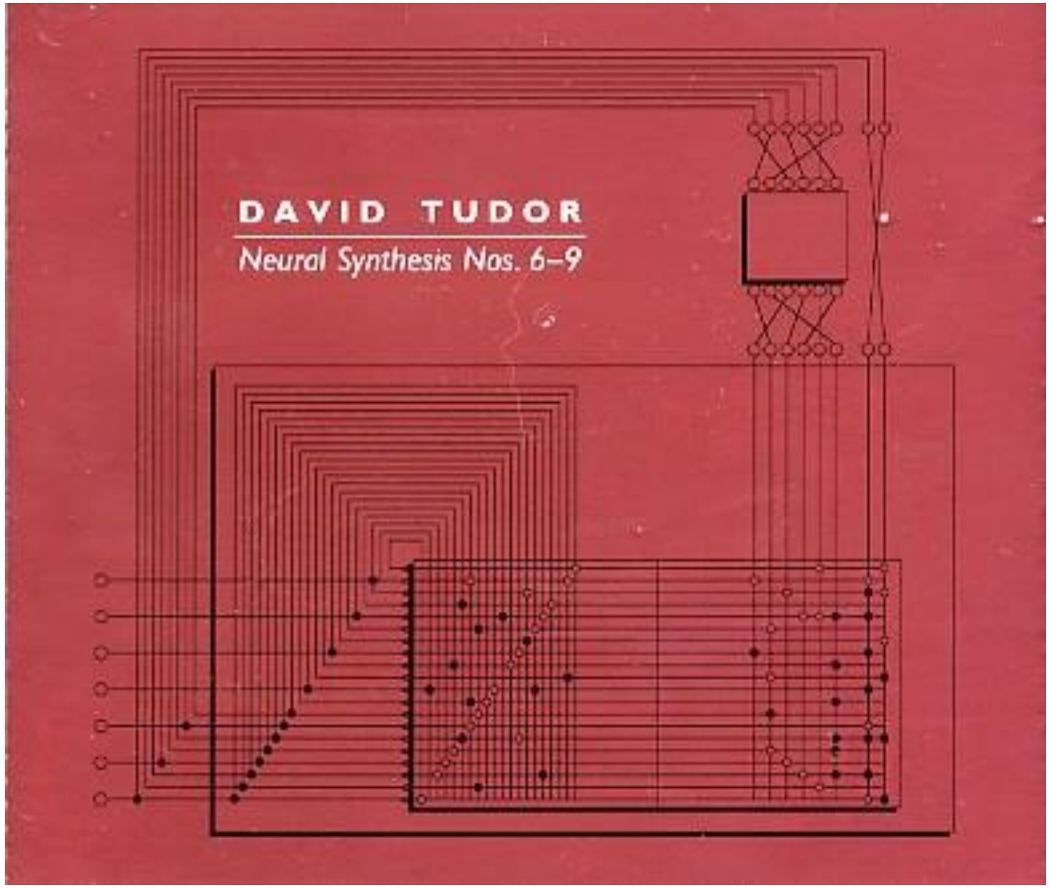
David Tudor and Intel, 1989-1993

'…because the Intel corporation was pleased by the fact that I had used this to make a product that was of interest to the company, and they were pleased enough so that they gave me the processor which weights the chips' (David Tudor, STEIM Archives 1994).

David Tudor and Intel, 1989-1993

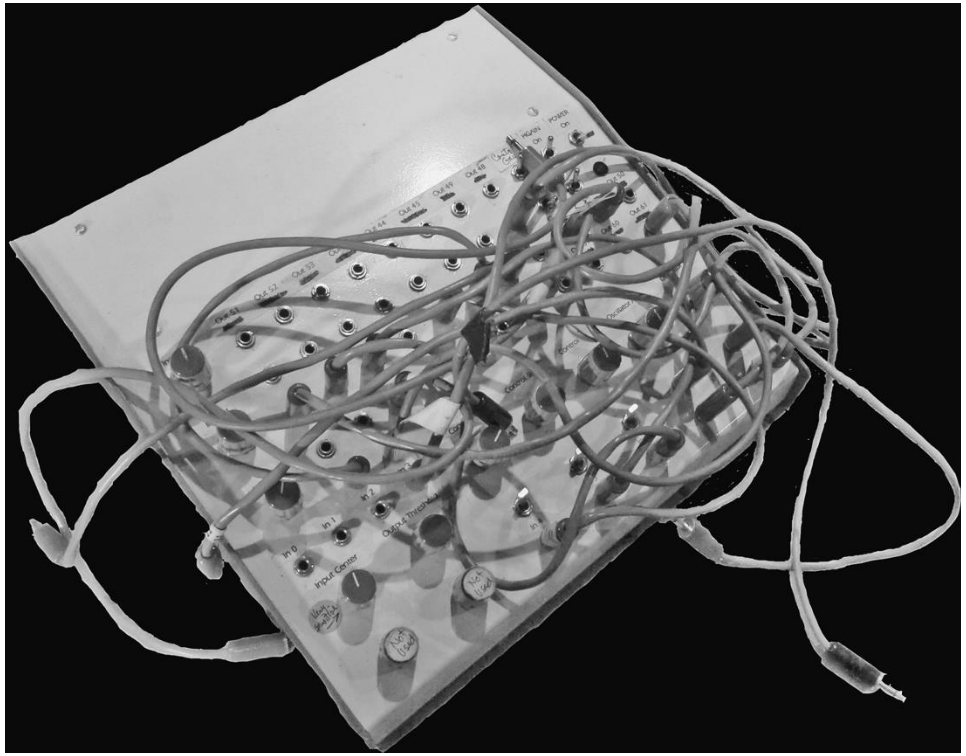


Signal flow diagram for Neural Network Synthesizer box 1, 1991



Cover for Neural Synthesis Nos. 6-9 (Lovely Music, 1995)

David Tudor and Intel, 1989-1993



Thorson, Warthman, and Holler, *Neural Network Synthesizer* DTC, Instrument 0459 | World Instrument Collection, Wesleyan University (from You Nakai 2021)

L-R Mark Holler, Mark Thornston, John Cage, and Forrest Warthman

David Tudor and Intel, 1989-1993

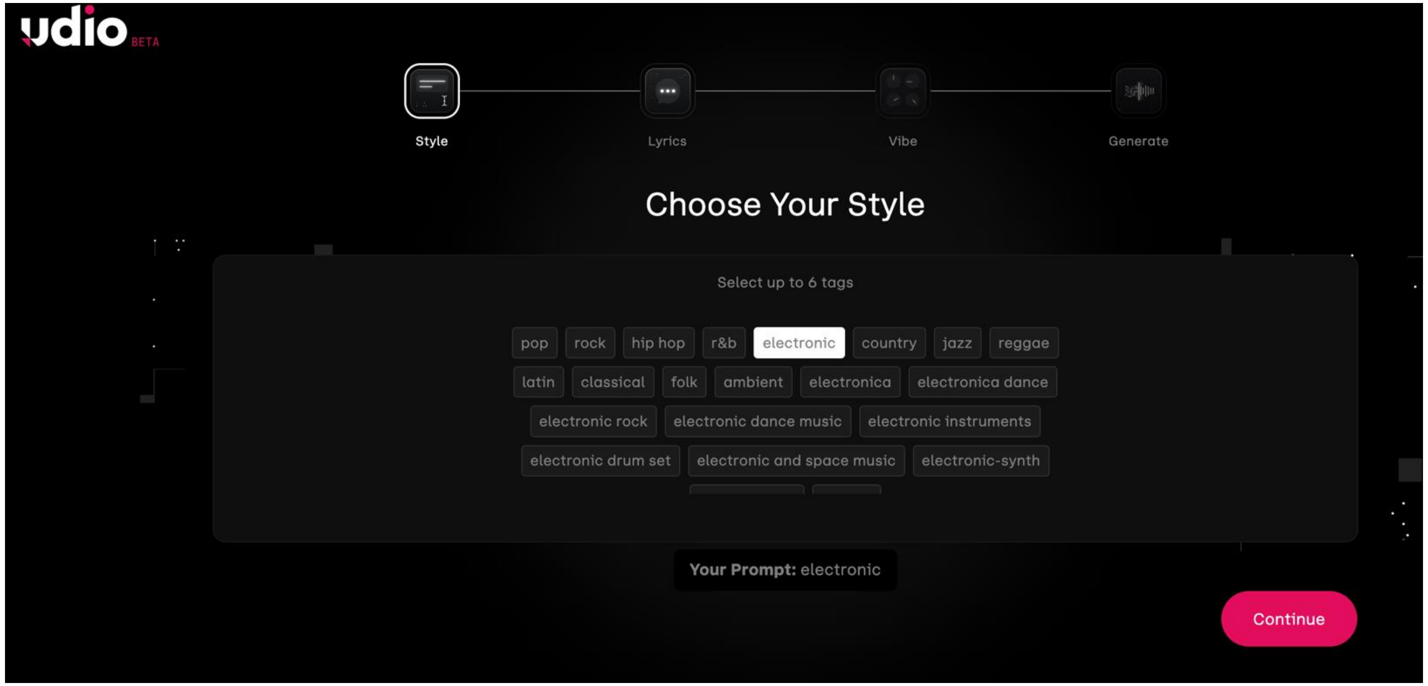
David Tudor, Neural Synthesis performance, Location: Steim Slaapkamer Muziek, 1994.

Critique?

Simondon Gilbert 1958 Le processus de concretisation. Via https://monoskop.org.

Critique?

Larry Levine and Phil Spector, Gold Star control room



Interface for Udio (beta), gen-ai studio

Conclusions

Changing fortunes of AI over its 70-year life have had an indirect but consequential impact on experimental music, especially in America but also in Europe (e.g. IRCAM)

This impact is at once political-economic and intellectual. Understanding the intellectual horizon against which Tudor, Amacher (and others) worked is key to understanding why they took the positions they did

For Amacher: critique of the human supremacist assumptions of AI; for Tudor, critique of AI’s aspirations to concrescence

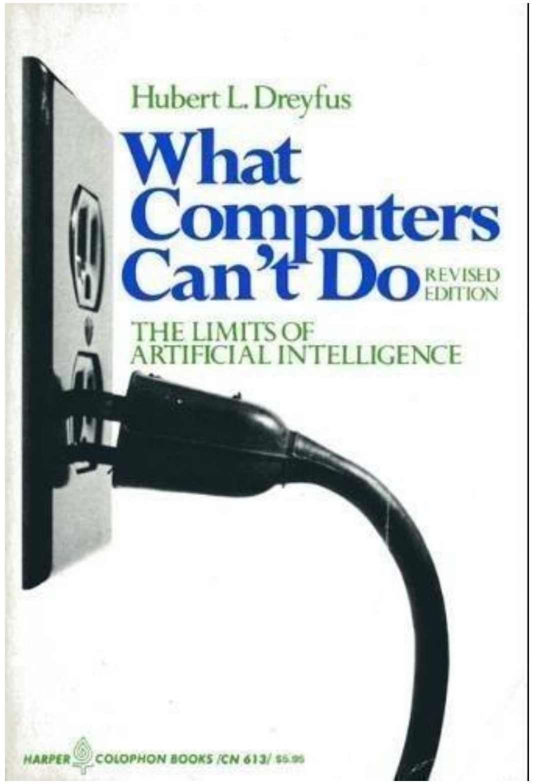
Questions

What are the political possibilities (and limits) of ‘artistic critique’? How should we understand it in relation to the broader tradition of AI critique in the humanities and social sciences?

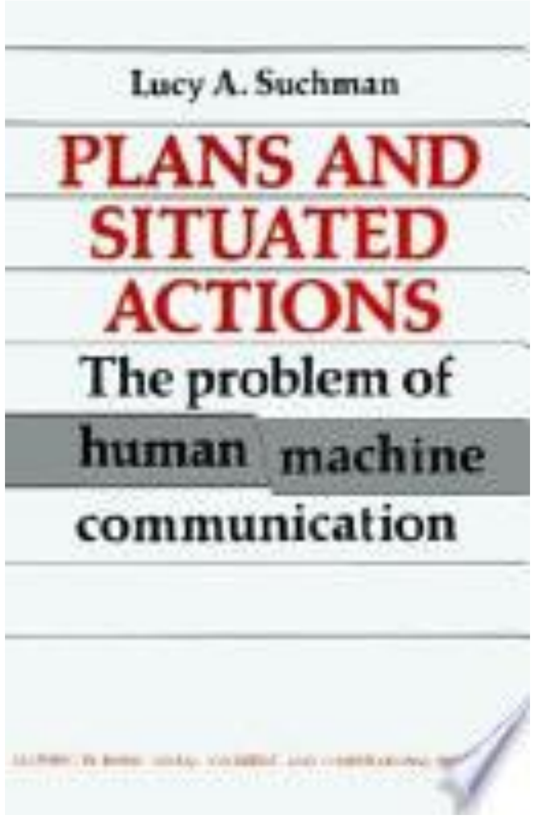
Where are the openings in today’s ‘inhospitable institutional and economic conditions’, inside which artists might approach contemporary AI with a ‘sense of possibility’?

What factors might make the third AI winter different?

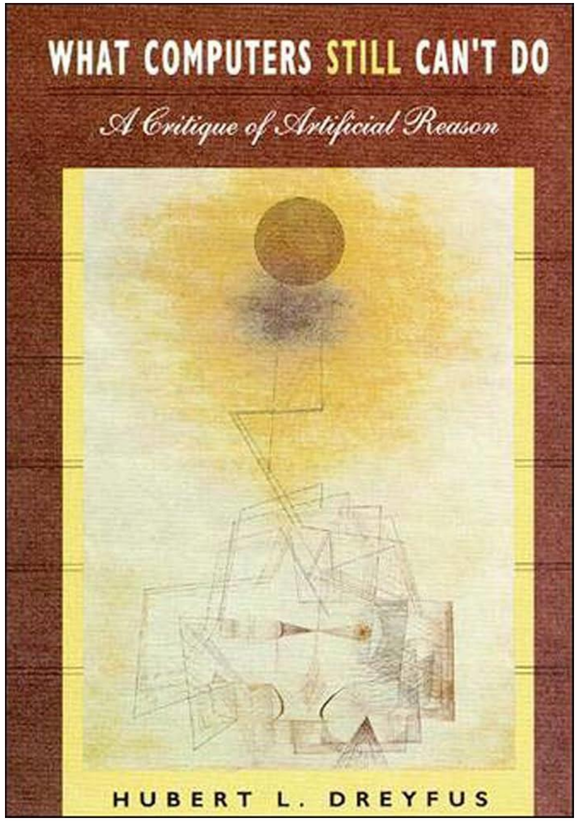
AI Critique in the humanities and social sciences



1972

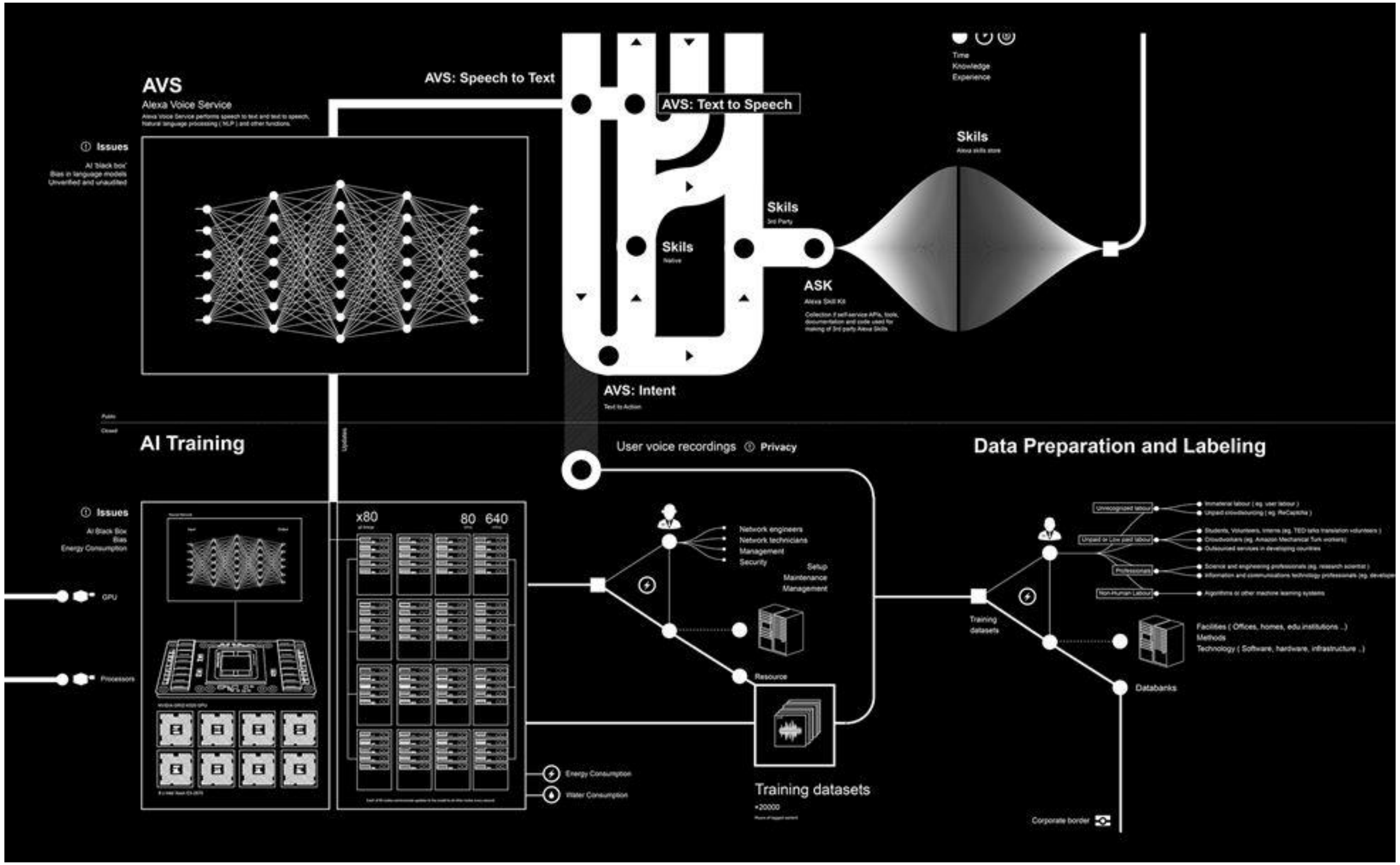


1987



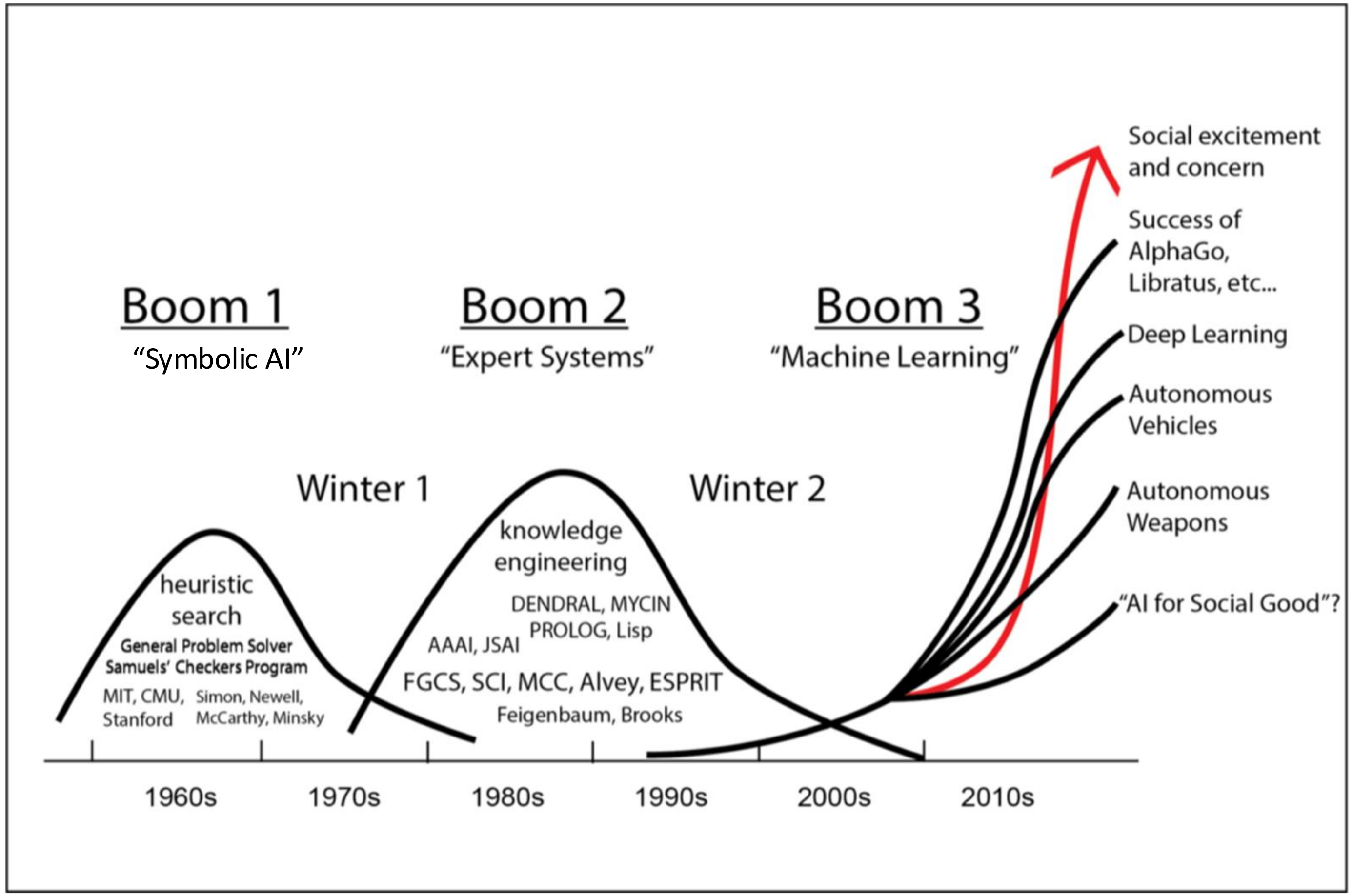
1992

AI Critique in the humanities and social sciences



Kate Crawford and Vladen Joler (2018) ‘Anatomy of an AI System’

What critical music might follow the *next* AI winter?



Adam, A., 1998. *Artificial Knowing: Gender and the Thinking Machine*. New York: Routledge.

Amacher, M. 1982 (approx). 'Dynamic Range / "The Head Stretch"'. Unpublished text. Maryanne Amacher Papers, The New York Public Library.

Bender E, Gebru T, McMillan-Major A, et al. (2021) On the dangers of stochastic parrots: Can language models be too big? FAccT' 21. https://doi.org/10.1145/3442188.3445922 (accessed May 2023)

Born, G., 1995. *Rationalizing culture: IRCAM, Boulez, and the institutionalization of the musical avant-garde*. Univ of California Press.

Cimini, A., 2022. *Wild Sound: Maryanne Amacher and the Tenses of Audible Life*. Oxford University Press.

Collins, H.M., 1990. *Artificial experts: Social knowledge and intelligent machines*. MIT press.

Collins, N., 2004. Introduction: Composers inside electronics: Music afterdavid tudor. *Leonardo Music Journal*, pp.1-3.

Crevier, D., 1993. *AI: the tumultuous history of the search for artificial intelligence*. Basic Books, Inc..

Dreyfus, H.L., 1972. What computers can't do: The limits of artificial intelligence.

Dreyfus, H.L., 1992. *What computers still can't do: A critique of artificial reason*. MIT press.

Edwards, P.N., 1996. *The closed world: Computers and the politics of discourse in Cold War America*. MIT press.

Flamm, K., 1988. *Creating the computer: government, industry, and high technology*. Brookings Institution Press.

Fredkin, E. and Minsky, M. 1970. Triadex Muse User Manual. Triadex Incorporated.

Fredkin, E. and Minsky, M. 1970. Digital Music Synthesizer. Patent US3610801A.

Garfinkel, S. 2021. 'Tomorrow's Computer, Yesterday'. MIT Technology Review. https://www.technologyreview.com/2021/04/27/1021714/tomorrows-computer-yesterday/

Gebru, T. and Torres, É.P., 2024. The TESCREAL bundle: Eugenics and the promise of utopia through artificial general intelligence. *First Monday*.

Haigh, T., 2023. There Was No 'First AI Winter'. *Communications of the ACM*, *66*(12), pp.35-39.

Haworth, C. 2023. 'Interview with Forrest Warthman and Mark Holler'. Zoom interview. 1st March 2023.

Hendler, J., 2008. Avoiding another AI winter. *IEEE Intelligent Systems*, *23*(02), pp.2-4.

Higgins, H. and Kahn, D. eds., 2012. *Mainframe experimentalism: Early computing and the foundations of the digital arts*. Univ of California Press.

Holler, M. Tam, Castro and Benson, . 1989. An electrically trainable artificial neural network (ETANN) with 10240 'floating gate' synapses. In *International 1989 Joint Conference on Neural Networks* (pp. 191-196). IEEE.

Hopfield, J.J. 1988. "On the effectiveness of neural network hardware". Proceedings of the workshop, *Hardware implementation of neuron nets and synapses*, San Diego (USA), P. Mueller ed, pp. 120–124, January 1988.

Intel Computers. 1991 (approx). *Neural Network Solutions*. Intel Product Catalogue. Consulted at: Getty Research Institute.

Kendall, G.S., Haworth, C. and Cádiz, R.F., 2014. Sound synthesis with auditory distortion products. *Computer Music Journal*, *38*(4), pp.5-23.

Kitano, H. and Hendler, J.A., 1994. *Massively parallel artificial intelligence* (Vol. 11). Menlo Park, CA: Aaai Press.

McCorduck, P. and Cfe, C., 2004. *Machines who think: A personal inquiry into the history and prospects of artificial intelligence*. AK Peters/CRC Press.

McDermott, D., Waldrop, M.M., Chandrasekaran, B., McDermott, J. and Schank, R., 1985. The dark ages of AI: a panel discussionat AAAI-84. *AI Magazine*, *6*(3), pp.122-122.

Minsky, M. and Papert, S., 1969. *Perceptrons*. Cambridge, MA: MIT Press.

Minsky, M. 1981. 'Music, Mind, and Meaning'. Computer Music Journal, Fall 1981, Vol. 5, Number 3. *Computer Music Journal*, *5*(3).

Minsky, M.L. and Laske, O., 1992. A conversation with Marvin Minsky. *AI Magazine*, *13*(3), pp.31-31.

Nakai, Y., 2021. *Reminded by the Instruments: David Tudor's Music*. Oxford University Press.

Nelson, A.J., 2015. *The sound of innovation: Stanford and the computer music revolution*. MIT Press.

Norberg, A.L. 1989. 'An Interview with MARVIN L. MINSKY'. Conducted 1 November 1989, Cambridge, Massachusetts. Charles Babbage Institute, Center for the History of Information Processing, University of Minnesota, Minneapolis.

Penny, S., 2019. Enactive–performative perspectives on cognition and the arts. *AI & SOCIETY*, *34*(2), pp.243-249.

Pickering, A., 2010. *The cybernetic brain: Sketches of another future*. University of Chicago Press.

Plotnick, R., 2012. At the interface: The case of the electric push button, 1880–1923. *Technology and Culture*, *53*(4), pp.815-845.

Fano, RM. 1966. Project MAC Progress Report III, Massachusetts Institute of Technology, July 1965-July 1966.

Simon, H.A., and Newell, A., 1958. "Heuristic Problem Solving: The Next Advance in Operations Research," Operations Research, Vol. 6 (January-February 1958).

Simondon, G., 2024. *Du mode d'existence des objets techniques*. Flammarion.

Spicer, D. 2024. 'Neural Network Chip Joins the Collection'. Computer History Museum Blog. https://computerhistory.org/blog/neural-network-chip-joins-the-collection/

Stark, L. and Crawford, K., 2019. The work of art in the age of artificial intelligence: What artists can teach us about theethics of data practice. *Surveillance & Society*, *17*(3/4), pp.442-455

STEIM archives. 1994. 'David Tudor + Nick Collins Interview'. 16 September 1994. https://www.youtube.com/watch?v=SxiUHQy97TM&t=3s

Suchman, L.A., 1987. *Plans and situated actions: The problem of human-machine communication*. Cambridge university press.

Suchman, L.A., 2023. The uncontroversial 'thingness' of AI. *Big Data & Society*, *10*(2), p.20539517231206794.

Thorson, M., Warthman, F., and Holler, M. 1993. "A Neural- Network Audio Synthesizer," Dr. Dobb's Journal (February 1993), 50 – 64

Tudor, D. 1966. Programme notes for *Bandoneon!*. In: 9 Evenings of Theatre and Engineering (programme book). New York: Experiments in Art and Technology.

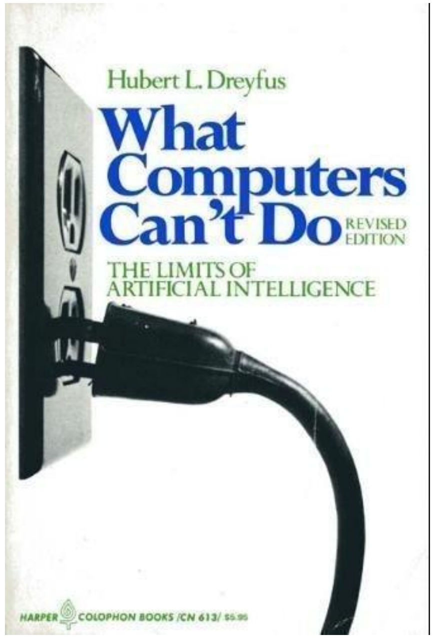
Valiquet, P., 2022. Grimes's Hymn to Technocracy, Insolvent Affordances, and the Need for Reparative Organology. *Journal of Popular Music Studies*, *34*(3), pp.119-141.

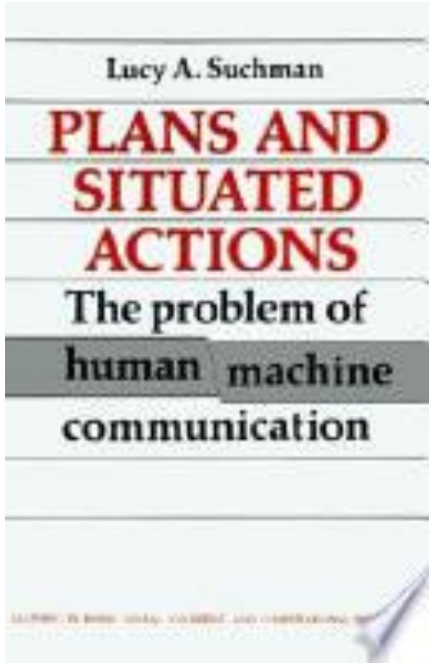
Van Rooij, I., Guest, O., Adolfi, F., de Haan, R., Kolokolova, A. and Rich, P., 2024. Reclaiming AI as a theoretical tool for cognitive science. *Computational Brain & Behavior*, *7*(4), pp.616-636.

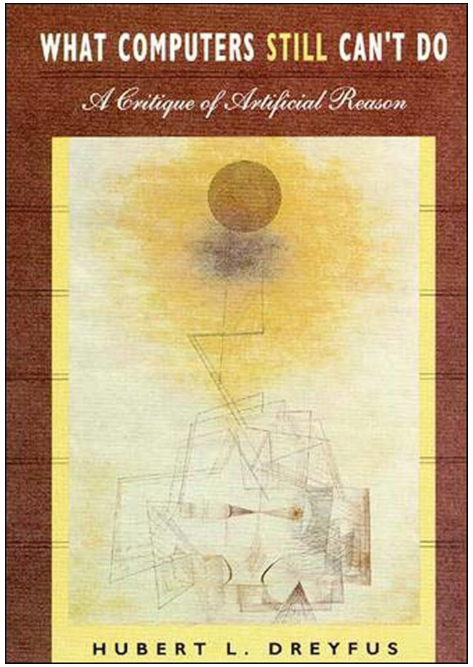
Warthman, F. 1995. "Liner Notes," David Tudor, Neural Synthesis Nos. 6- 9, Lovely Records (CD1602), 2CDs.

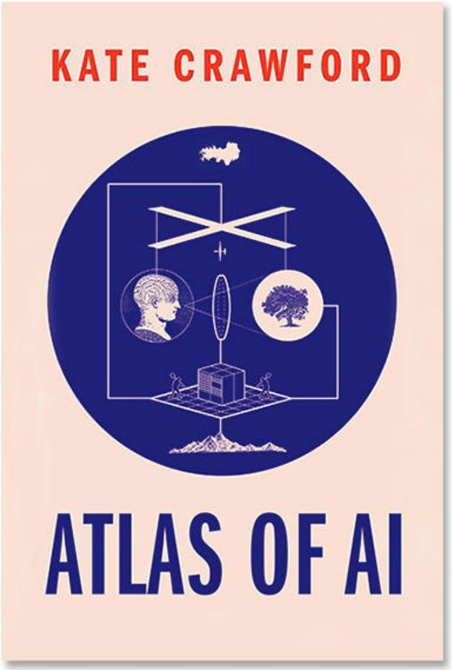
Warthman, F., Clapman, R., and Tudor, D. 1989. "A Performance Computer: Design Definition, Box 27, Folder 5, David Tudor Papers, GRI.

AI Critique









The TESCREAL bundle: Eugenics and the promise of utopia through artificial general intelligence

Timnit Gebru

Émile P. Torres

DOI: https://doi.org/10.5210/fm.v29i4.13636

Abstract

The stated goal of many organizations in the field of artificial intelligence (AI) is to develop artificial general intelligence (AGI), an imagined system with more intelligence than anything we have ever seen. Without seriously questioning whether such a system can and should be built, researchers are working to create "safe AGI" that is "beneficial for all of humanity." We argue that, unlike systems with specific applications which can be evaluated following standard engineering principles, undefined systems like "AGI" cannot be appropriately tested for safety. Why, then, is building AGI often framed as an unquestioned goal in the field of AI? In this paper, we argue that the normative framework that motivates much of this goal is rooted in the Anglo-American eugenics tradition of the twentieth century. As a result, many of the very same discriminatory attitudes that animated eugenicists in the past (*e.g.*, racism, xenophobia, classism, ableism, and sexism) remain widespread within the movement to build AGI, resulting in systems that harm marginalized groups and centralize power, while using the language of "safety" and "benefiting humanity" to evade accountability. We conclude by urging researchers to work on defined tasks for which we can develop safety protocols, rather than attempting to build a presumably all-knowing system such as AGI.

1972 | 1987 | 1992 | 2021 | 2024

What can artistic critique do?

“(Artists are at) **the vanguard of exploring the ethical, political, and aesthetic potential of emerging technologies and in recognizing the interconnection between the techniques of artistic production and their political effects** (…) society at large will be increasingly called on to make the kinds of judgments—aesthetic, ethical, and political—that the artists we interviewed were simultaneously exploring, critiquing, and struggling with (Stark and Crawford 2019, 451, 452)

Artists engaging with data analytic tools and computational techniques can share their opinions on the ethical importance, impact, and implications of their projects, **often without being beholden to particular companies or platforms**. This makes them a significant but often undervalued community able to speak directly to the changing and nuanced ethical questions faced by those who use data, machine learning (ML), and AI in their work.” (443)

Stark, Luke, and Kate Crawford. 2019. The Work of Art in the Age of Artificial Intelligence: What Artists Can Teach Us About the Ethics of Data Practice. Surveillance & Society 17(3/4): 442-455.